



Anti-snore wearable

This wearable automates a poke to make you stop snoring in bed and make you turn to the side <http://dgit.in/antiSnr>



Microsoft accused

Kaspersky founder says pre-bundling Defender in Win10 is anticompetitive. <http://dgit.in/ksprskMS>

Touched by tech

Similarly, for ECG, Alivecor manufacturers the Kardia range of products. The Kardia mobile is a \$99 device that can get medical grade ECG analysis in under 30 seconds, while being fit right into a smartphone case. And like most devices these days, it works with an app to relay the data that you can share with your physician. They are also planning to launch Kardia band for the Apple Watch, although that has not been cleared by the FDA yet.

Multiple other devices like the Tinke from Zensorium and Withings lineup of products that are also equipped with multiple sensors that detect things like blood Oxygen saturation, heart rate variability, stress indices and more.

Highly advanced prosthetics

The bebionic prosthetic arm claims to be the world's most lifelike prosthetic arm. It was launched way back in 2010 and has advanced a lot since then. It uses microprocessors and individual motors to simulate 14 life-like grip actions for opening doors, eating food and more. It even comes in multiple sizes to accommodate for different age groups and body structures.

On the other hand, Japanese company SHC design aims to solve the problem of prosthetic limbs being expensive and non- customisable by bringing 3D printing into the process. While average costs of a prosthetic limb is at least around \$5000, 3D printing them could bring it down to as low as \$100 per limb, according to SHC Design Chief Operating Officer Yutaka Tokushima. And they would also be highly customisable and would be able to get the proportions more accurately, even account for certain accessories and clothes.

And then there are so many examples of high-tech one-off prosthetic limb designs that have pushed the boundaries of what is possible in prosthetics – from thought controlled limbs to video-game inspired designs. Overall, the future does look pretty bright for this area.

Telemedicine and remote surgery

Internet brings connectivity. What comes with connectivity is a host of services that were out of everyone's immediate reach earlier. Now you can pick up your phone and order a cab, get food delivered to your house and even visit a doctor. Telemedicine, or the provision of healthcare from a distance using telecommunications and information technology, has been around for a while. But right now, in our nation, we are witnessing a widespread availability of affordable 4G and smartphones. Those are essentially the two things needed to get a telemedicine project up and running and that's where apps like DrLive come in.

"All over the world, there is widespread acceptance of telemedicine for its usefulness to impact routine care for common, uncomplicated, non-emergency medical problems related to sinus, respiratory infection, aller-

ment to an actual clinic visit for patients who can use it to for pre-visit and post-visit consults which would help them reduce a lot of travel related costs and benefit from the added convenience."

Medicine delivery

For ages, we've been consuming medicine the same way. Even though the medicine itself has evolved over the years into compact pills and capsules, which target pretty specific ailments and areas (compared to the broad spectrum treatments of yesteryear) there is still a significant degree of difficulty to access certain areas through traditional drug delivery methods. Advancements in nanotechnology are changing that.

"Semiconductors are typically brittle and rigid, but once they are thinned and oriented down to the micro and nano-scale, they become extremely soft and bendable. These unique mechanical properties of Silicon nano-membranes and other thin-film electronics underscore the enormous potential of nanotechnology to play a role in healthcare. You can imagine designing devices and networks of biosensors based on these silicon nanomembranes and thin-films, interfacing with soft tissues and cells for diagnostic and therapeutic applications", says Rooz Ghaf-fari, Nanotechnologist, Co-Founder-MC10 and speaker at TEDx Gateway 2016 this year at Mumbai, 2016.

He goes on to explain the potential in further detail. "In drug delivery and therapeutic-based applications, nano-particles are used to deliver molecularly targeted agents to tumours, thereby reducing toxicity side effects associated with systemic delivery of drugs. In these cases, nano-particles engineered to provide high specificity, are attractive as auto-immune disease targets and cancer therapy." Even on other sides, nanotechnology aims to make healthcare's reach wider and more specific at the same time. Overall, closed loop systems that deliver targeted nano-particles and measures the effects of the drug using ultrathin sensors would be a major breakthrough in healthcare. 



3D printed prosthetic limbs could cost way too less

gies, urinary tract infection, cold and flu symptoms and many other non-emergency illnesses", says Mr. Chethan Channakeshava, Co-founder, DrLive.

Even if you don't think that telemedicine could be a complete alternative to traditional doctor visits, it is evident that it is a very effective way to obtain healthcare that can be provided remotely without going through the hassle of appointments and visits. As Mr. Chethan puts it, "Through telemedicine, DrLive enables secure audio-video medical consults with a mission to emerge as a new frontier in telemedicine – one that offers safe, convenient, and cost-effective options for users. Telemedicine can therefore be a great compli-